Ray Tracing

> SYNOPSIS
>
> Note 6 found the image by arithmetic. This note finds it with a straightedge. Three rules — one for a ray parallel to the axis, one for a ray through the middle, one for a ray through the focus — are enough to locate any image a thin lens can make, and to see at a glance whether it is real or virtual.

BEFORE YOU START — PREDICT

You trace two rays carefully and they **spread apart** after the lens instead of crossing. Have you made a mistake, or has the lens told you something? Decide now.

PERFORMANCE EXPECTATIONS

1. **State** all three ray rules without looking them up.
2. **Construct** an image by tracing at least two rays from the tip of the object.
3. **Distinguish** a real image from a virtual one by whether the refracted rays themselves cross, or only their backward extensions do.
4. **Check** a construction against $1/d_o + 1/d_i = 1/f$, and say what it means when they disagree.

3. Ray tracing

Two names for the same line, and both are used: the **optic axis** and the **principal axis**.

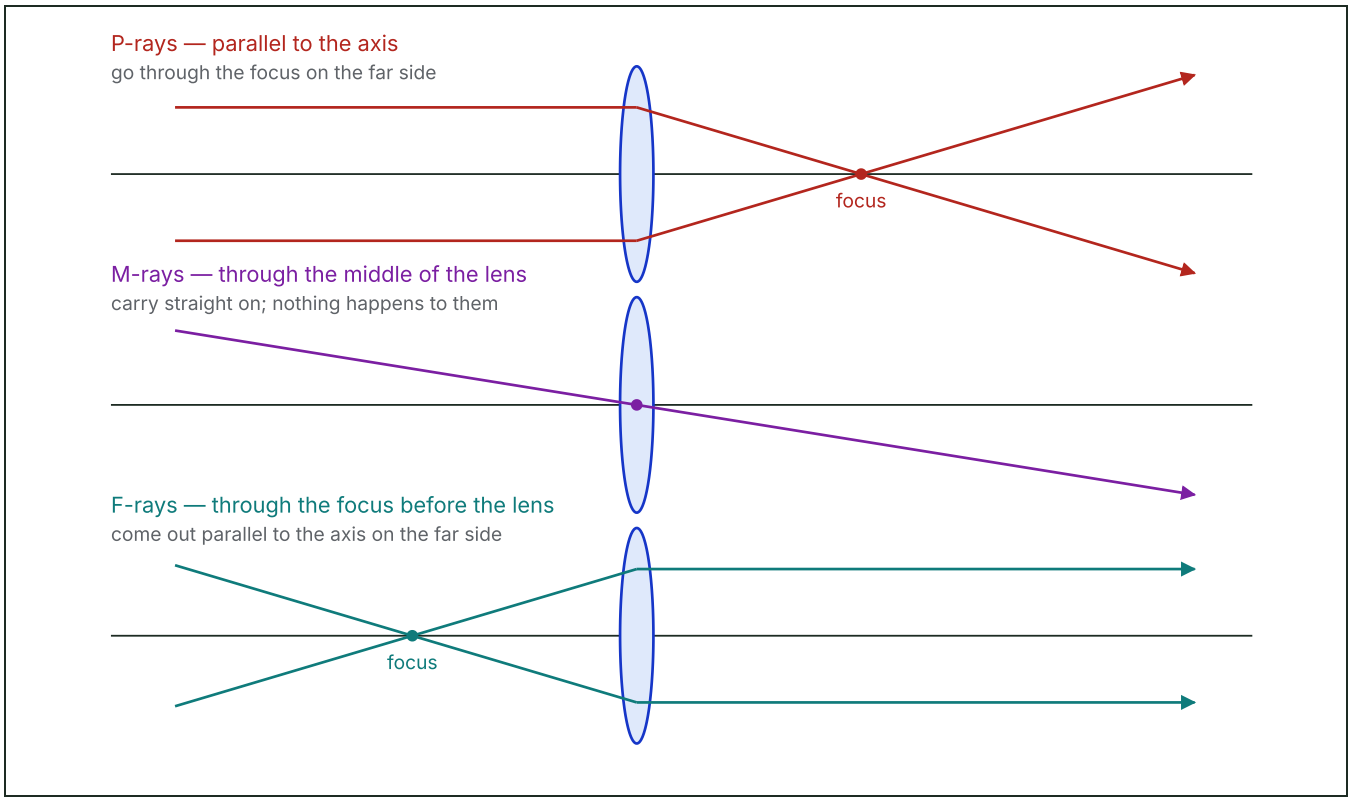


Figure 1. Three rules, and that is the whole of ray tracing. Each one says what a lens does to *one particular* ray — and every one of the three is reversible, which is why you may trace in either direction.

Your job on Figure 2. Figure 1 above gives you the three rules. Apply two of them here: the object sits *inside* the focal length, so carry both incident rays through the lens, then extend them **backwards** until they meet, and draw the image you find there. Is it real or virtual?

- **Rays parallel to the principal axis** ⇒ go through the focus on the other side of the lens.
- **Rays going through the centre of the lens** ⇒ nothing happens; they keep going without bending.
- **Rays travelling through the focus before the lens** ⇒ come out parallel to the optic axis on the other side.

Any **two** of the three locate the image. The third is the check — if it does not arrive at the same point, something in the construction is wrong, and it is nearly always the focal point that has been put on the wrong side.

Worked example — trace it with no numbers at all

1. An object stands **outside** the focal length of a converging lens. Using nothing but the three rules above — no distances, no equation — locate its image, and say whether it is real or virtual, upright or inverted, before you measure anything.

When the rays refuse to cross

Move the object *inside* the focal length and the construction changes character. The refracted rays still obey all three rules, but they diverge.

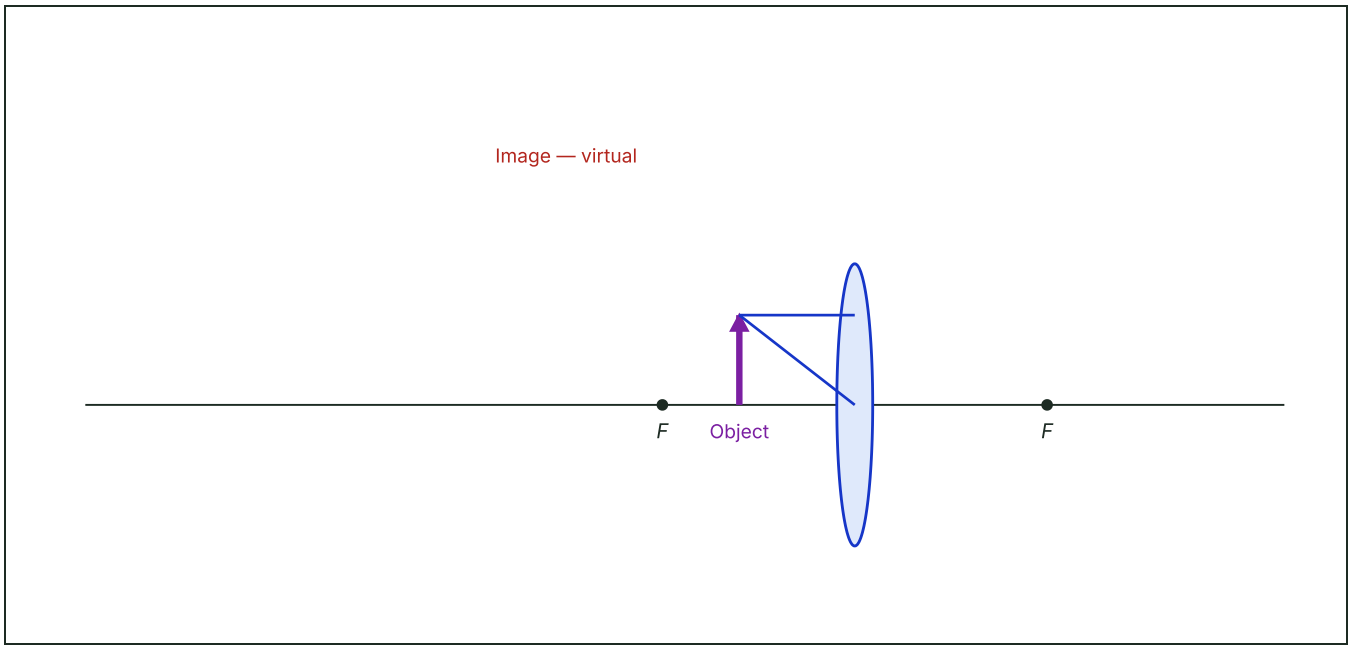


Figure 2. Object inside the focal length. The two refracted rays never cross, so they are traced *backwards* (dashed) until they do. That meeting point is the virtual image — upright, magnified, and on the same side as the object. The arithmetic agrees: d_i = −225 px and m = +2.5.

Trace them backwards and they meet. That meeting point is a **virtual image**: no light actually passes through it, which is why you cannot catch it on a screen — but your eye, which only ever sees the directions rays arrive from, sees an object standing there.

THE PUNCHLINE

Parallel → through F · through the middle → straight on · through F → parallel

Back to the prediction: rays that spread apart are not a mistake. They are the lens telling you the image is virtual — the same message the equation sends as a negative d_i.

Looking ahead

Three rules and one equation now cover every single thin lens. **Note 8** keeps them and adds curved mirrors, which turn out to need no new physics at all — only a sign convention of their own.

2. Which ray is unchanged by a thin lens?

○ The one parallel to the axis. ○ The one through the near focal point. ○ The one aimed at the centre of the lens. ○ The one arriving at 45° to the axis.

3. In a finished construction, the refracted rays themselves never meet — only their backward extensions do. The image is:

○ Real, and could be caught on a screen. ○ Virtual, and could not be caught on a screen. ○ Nonexistent — no image is formed. ○ Real, but inverted rather than upright.

4. How many of the three rays do you need to locate an image?

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.